

Raphael Catenacci

Montréal – Canada

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Education

Université de Montréal <i>M.Sc. Exchange in Mathematical and Computational Finance</i> ○ One-year academic exchange as part of the ENSAE Paris engineering curriculum.	Montréal, Canada 2026 – 2027
ENSAE Paris <i>Engineering School of the Institut Polytechnique de Paris</i> ○ Relevant Courses: Financials Mathematics, Stochastic Processes, Machine Learning Fundamentals, Financial Instruments, Time Series, Statistical Simulations and Monte Carlo Methods, Python for Data Science (Scikit-Learn, Geopandas, API and Webscrapping).	Palaiseau, France 2023 – 2027
International Center of Valbonne <i>Mathematics and Physics Preparatory Classes</i>	Sophia Antipolis, France 2021 – 2023

Professional Experience

Jukoï Capital <i>Quant Researcher off-cycle</i> ○ Developed a fully automated, provider-agnostic financial data infrastructure for scalable market data ingestion and processing. Leveraging local LLMs to simplify and structure macroeconomic and alternative data ingestion workflows. (Languages: Python, JavaScript, HTML) ○ Implemented sampling, meta-labeling, and sample-weighting methodologies on datasets to identify stationary-memory optimal solution and maximize predictive signal power prior to machine learning model deployment. ○ Automated the financial instrument lifecycle management process, including corporate actions handling, pricing updates, event processing, and portfolio synchronization. ○ Implemented risk calculations, stress testing, exposure measurement, and pricing models for trading portfolios, with seamless integration into a web-based financial application.	Monaco Jun 2025 – July 2026
AXA France <i>Data Scientist Intern</i> ○ Developed algorithms within the financial compliance team. ○ Built an artificial neural network using the Transformers library and utilisation of Azure Cloud. Used Pandas, Scikit-Learn, and Pytorch for Machine Learning.	Paris Jun – Sep 2024

Skills

Programming: Python (Advanced), JavaScript, HTML, CSS, SQL, R, SAS, VBA
Software: Microsoft Office, LaTeX
Languages: French (native), English (Fluent) – MaltaLingua School of English certificate

Projects

2025: Applied Statistics project for ENGIE: Developed a CatBOOST Machine Learning model to forecast French national gas consumption. Performed fine-tuning of hyperparameters and leveraged AutoML tools to optimize model performance.

2025: Financial Machine Learning Automatic Trading Algorithm: Designed and implemented various trading strategies using machine learning techniques. Conducted performance evaluation with the calculation of key financial metrics such as Sharpe Ratios, drawdowns, and other risk-adjusted performance indicators.

2024: Real Estate Price forecast: Forecasting real estate prices using Machine Learning algorithms and public datasets.

2024: Python Project: Optimizing routes by matrix arrangement using the A-star algorithm.

2023: Economics Thesis: Analyzing the impacts of the pension system on the French economy.

2023: Statistics Thesis: Data study on French wealth distribution using SAS.

2022–2023: TIPE: Developed a functional Li-Fi communication system.

Extra-Curricular Activities

Sports: BMX Racing (2019 UCI World Ranking U18: 9th)
Mathematics: Participation in the 2021 French general mathematics competition
Associations: Member of ENSAE Junior Enterprise, President of ENSAE Sports Office in 2024-2025